

Predicting School Enrollment – R Data Science Project

Instructions: You MUST follow the format listed below and use the techniques as shown below (e.g. panelr for transposing). It is okay to add R-code steps and/or syntax, but do not change, avoid, or disregard any of the listed steps (without permission). Important: You are not allowed to use bracket-column number syntax because that process cannot be scientifically replicated. For example, file1[,5:15] is not allowed (except for in the final data analysis, e.g. correlations, running regressions). Be sure to include comments in your R code.

READ-IN AND PREPARE THE SCHOOL ENROLLMENTS DATA

1. At the top of your R code file, read-in most of the libraries that you will need.

For example: `library(tidyverse), library(panelr)`.

2. We will work on the school enrollment data first. Read-in ELSI_School_Enroll_1990_2001_Revised.xlsx. Create an exact copy called "school_clean1"

3. Prepare/clean the file prior to transposing and merging. To replicate the SAS process, write some code to replace all "dagger" ("†") values and dashes ("–") from all variables. Reassign the values to NA.

Base R is easiest for this task:

```
school_clean1[school_clean1=="†"] <- NA #replace all "dagger" values with NA
```

4. Change all variable names to lowercase.

5. Transpose "school_clean1" from wide to long panel format by using the `long_panel` command (from `panelr`). Be sure that you understand the code example pasted below.

Code example:

```
school_long <- long_panel(school_clean1, id="school_id", prefix="_", begin = 1990, end = 2001, label_location = "end")
```

6. After the transpose, rename the variable "wave" (created by `long_panel`) to "year" and change "white", "black", "hispanic" and all other race/ethnic groups from character into numeric variable types.

7. Using `dplyr` and the `rowwise()` command, create a variable called "total_enroll2" by summing all of the race/ethnic counts together for each school. Sort the data by `school_id`, year to inspect your enrollment counts and look over the data (some missing values are expected). Drop the original "total_enroll" variable as it is no longer needed.

8. Our `school_id` variable is now a "factor." Using base R and `dollar$` syntax, convert `school_id` to numeric. Note: This can be tricky because you may have to convert to character first, then numeric (all on a single line of code).

READ-IN AND PREPARE THE SCHOOL COUNTY FIPS DATA – MERGE WITH ENROLLMENTS

9. We now need a unique variable for each county (e.g. `county_fips`). Read-in both school files that contain county fips using these names: **`school_counties`** <- ELSI_School_ID_County_FIPS_2001.xlsx and **`school_coord`** <-

All_School_Coordinates_2016_ELSI.xlsx. Be sure to convert `school_id` in both of these files into a numeric or integer format for merging with `school_enrollment` (using a single line of Base-R code right after the read-in may work best.)

10. Prepare / clean both files that contain `fips_county` (one at a time). Lowercase all variables and keep only the needed variables when merging. If needed, fix the improper two-word "school name" variable using this code: `rename(school_name2="school name") %>%`

11. Use `dplyr` left join to merge the enrollments data frame with **`school_counties`** (create: **`combo_school_counties`**). Then, create a data frame with ONLY the non-matching records from the merge. Merge this non-matching file with **`school_coord`** using an `inner_join`. Lastly, merge the matching records from this merge with `combo_school_counties`.

12. Clean the final enrollments (combo) file by removing unnecessary variables. Use coalesce to ensure that the fips_county variable is as complete as possible.
13. Aggregate the final enrollments (combo) file to the county-year level by using:
summarize(county_enroll=sum(total_enroll2)) with groupby. You are done with the school enrollment file.
14. Create a permanent csv for the county-year level school enrollments file that you just created and write syntax to clear all files from the R Global Environment using: rm(list=ls()). Next, read-in the csv that you just created to start fresh.

READ-IN AND PREPARE THE CENSUS AND CRIME DATA – MERGE WITH SCHOOL ENROLLMENTS

15. Use library(haven) to read-in **census_orig** using the SAS dataset "pop_est9001.sas7bdat". Clean and prepare this file for merging by creating a numeric or integer fips_county variable and removing unnecessary variables. To aid in cleaning/formatting, consider this code: mutate(population=pop, county_name2=county, state_name=state) %>%
16. We now need census population counts by race and age for use as predictors. Use dplyr left_join to merge the aggregated (county-year) level school enrollment file with the county-year level census file.
17. Read-in the SAS agency-year level crime data "ucr_1990_2001_ori_level.sas7bdat" and prepare this file for merging with enrollments.
18. You will need to create a unique concatenated fips_state, fips county (i.e. fips_state | fips_county) variable for the merge with enrollments. To combine fips_state and fips_county together (in char format with leading zeros), use:
mutate(fcounty=(formatC(fips_county, digits = 0, width=3, format="d", flag="0"))) %>% Repeat this process for fips_state. Then use str_c or "string concatenate" function to combine the two fips variables together as fips_county.
19. Use dplyr summarize with groupby to aggregate the crime data to the county-year level. Be sure to keep the key crime variables:
total_crime_county=mean(total_sum),
total_crime_rate_old=mean(total_rate),
violent_crime_county=mean(violent_sum),
violent_rate_old=mean(violent_rate)) %>%
20. Use dplyr left join to merge the combo_school_enrollments data (from step 16) with the aggregated crime data.

READ-IN AND PREPARE THE ECONOMIC DATA – MERGE WITH ENROLLMENTS

21. Read-in and prepare the economic files using the following code:
library(haven)
files = list.files(path="J:/Your Path Name/BEA_Economic/",pattern="*.sas7bdat", full.names=TRUE)
j=1989
for (i in files) {
 j=j+1
 assign(paste0("reis_",j),read_sas(i))
 temp=get(paste0("reis_",j)) #creates temp file for renaming variables and adding year
 names(temp)<- c("fips_county","area_name","state","percap_income","employment")
 temp\$year=j
 assign(paste0("reis_",j), temp) #Renames temp back to original reis_yyyy
}
econ_vect <- mget(ls(pattern="reis_"))
econ_orig <- bind_rows(econ_vect) #Stacks the econ files together

22. Convert fips_county to numeric or integer format. Then, left_join the most updated enrollment file with the economic variables.

23. Create two new county-wide crime rate variables:

```
mutate(new_tot_crime_rate=(total_crime_county/population)*100000) %>%
```

```
mutate(new_violent_rate=(violent_crime_county/population)*100000) %>%
```

Also, create an employment rate variable: `mutate(employment_rate=(employment/population*1000))`

24. Finish cleaning and relocate variables as needed. For example: `relocate(new_tot_crime_rate, new_violent_rate, .before=area_name)`. You now have your final merged file.

25. Demean the independent and dependent variables for use in panel data regressions:

```
school_dm_final <- school_dm_final1 %>%
```

```
group_by(fips_county) %>%
```

```
mutate(across(c(county_enroll:employment_rate), ~.x - mean(.x), .names = "{col}_DM"))
```

26. Write this final file with DM variables into a permanent csv file. You may again wish to clear your Global Environment, read-in the permanent csv that you just created and start fresh. Analytics is next.

GET ANSWERS – DATA ANALYTICS

27. Create nation-year descriptive variables to check overall trends. Were enrollments, crime, econ up or down over the 12-year time period (1990-2001)? A few simple line plots are optional.

```
national_schools1 <- school_dm_final %>%
```

```
group_by(year) %>%
```

```
summarize(county_enroll_dm=mean(county_enroll_DM, na.rm=TRUE),
```

```
prop_nonwhite_dm=mean(prop_nonwhite_DM, na.rm=TRUE),
```

```
prop_age1524_dm=mean(prop_age1524_DM, na.rm=TRUE),
```

```
total_crime_rtdm=mean(total_crime_rate_old_DM, na.rm=TRUE),
```

```
employment_rate_dm=mean(employment_rate_DM, na.rm=TRUE),
```

```
percap_income_dm=mean(percap_income_DM, na.rm=TRUE),
```

```
violent_rate_old_dm=mean(violent_rate_old_DM, na.rm=TRUE))
```

Optional line plot code (you may also use ggplot2):

Not required: `plot(national_schools1$year, national_schools1$county_enroll_DM, type = "l")`

28. Check inter-variable correlations (any values higher than .65 should be used with caution)

```
library(corr)
```

#To check the first half of the variables

```
corr1 <- school_dm_final %>% #Creates a nice, clean correlation dataframe.
```

```
correlate() %>% # Create correlation data frame (cor_df)
```

```
focus(county_enroll_DM, population_DM, prop_nonwhite_DM, prop_male_DM, prop_age1524_DM, mirror = TRUE) %>%
```

```
rearrange() %>% # rearrange by correlations
```

```
shave() # Shave off the upper triangle for a clean result
```

#To check the second half of the variables

```
corr2 <- school_dm_final %>% #Creates a nice, clean correlation dataframe.
```

```
correlate() %>% # Create correlation data frame (cor_df)
```

```
focus(county_enroll_DM, violent_rate_old_DM, new_violent_rate_DM, percap_income_DM, employment_rate_DM, mirror = TRUE) %>% # Focus on cor_df without 'cyl' and 'vs'
```

```
rearrange() %>% # rearrange by correlations
```

```
shave() # Shave off the upper triangle for a clean result
```

Question: What variable is most correlated with “year”?

29. Regress enrollment by each predictor, one at a time. Lastly, run the final regression model with all statistically significant predictors together.

30. Which census, economic, and crime variables significantly predicted county school enrollment from 1990-2012? What was the overall predictive value of the final regression model (adjusted R-square)? Which variables predicted a decline in enrollment (e.g. people leaving the area due to bad conditions in the county)? In a few brief sentences, explain your results.